RTX 4080 SUPER

Physical VRAM, tensor-rate and lock audit

Read-only BAR0 MMIO audit of the local AD103 card. The capacity figures come from the GPU's own registers, not from the display driver.

Report ID	GPU-HW-2026-11-07-4080S
Report date	2026-11-07 17:52 CET
Target	NVIDIA GeForce RTX 4080 SUPER, AD103, 10de:2702
PCI address	0000:04:00.0
VBIOS	95.03.44.00.4e
Board / GPU part	0x400 / 2702-400-A1
Host	Gigabyte Z790 GAMING X AX, BIOS F14, i9-14900K, 96 GiB DDR5
Method	ssdiag 0.6.14, signed pmxdrv.sys, BAR0 MMIO
Result	16 GiB physical; 16,376 MiB usable; tensor override live

Headline: 16,384 MiB raw · 16,376 MiB VBIOS-usable · 8 MiB VBIOS reserve · 329 MiB driver reserve · 16,047 MiB application-visible. There is no hidden 24 GB or 32 GB pool.

The tensor unlock is already live. This report is a read-only audit; it does not write FEAT, TPC, WPR or memory-partition registers.

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1. Executive summary

This card is a 16 GiB RTX 4080 SUPER. The physical GDDR6X population is 16,384 MiB. The VBIOS publishes 16,376 MiB as usable framebuffer, withholding 8 MiB. The Windows driver additionally reserves 329 MiB, leaving 16,047 MiB visible to applications. There is no hidden 24 GB or 32 GB pool behind a fuse, WPR window, or driver lock.

The tensor-rate override is currently active: `FEAT_PLM = 0xffffffff`, `SS0 = 0x88888888`, `SS1 = 0x00000008`, `READOUT_1 = 0x00000000`. The card has 40 active TPCs / 80 SMs / 320 tensor cores; GPC3 has 2 TPCs (4 SMs / 16 tensor cores) disabled by binning. That TPC cut is not the tensor-rate limiter and has no supported enable path.

Two platform limits matter more than any remaining GPU "lock":

- PCIe is electrically x4, not x16. The card is on the PCH root port `00:1c.4`; it trains Gen1 x4 at idle and can train Gen4 x4, but never x16 in this slot.
- BAR1 is 256 MiB of a 16 GiB-capable aperture. Re-Size BAR is not enabled. This is a CPU mapping window, not lost VRAM.

The 320 W power limit is a VBIOS power-table cap. It limits sustained tensor gains but is not a fuse and is not part of the tensor unlock.

1.1 Headline numbers

Question	Answer
Raw GDDR6X population	16,384 MiB (16 GiB)
VBIOS usable FB	16,376 MiB
VBIOS/hardware reserve	8 MiB (0.049%)
Driver/WDDM reserve at snapshot	329 MiB
Application-visible total	16,047 MiB (15.67 GiB)
Hidden/unlockable extra VRAM	0 MiB
Tensor-rate lock	Open / full rate
Active tensor hardware	40 TPC / 80 SM / 320 tensor cores
Disabled TPCs	2 TPC / 4 SM / 16 tensor cores in GPC3
CPU-visible VRAM aperture	256 MiB (BAR1; 16 GiB capable)
PCIe	Gen4 x16 capable, x4 electrical
Power limit	320 W
ECC	Off
MIG	N/A

2. Scope and method

2.1 Why direct BAR0 reads

`nvidia-smi` and NVAPI report the driver's view. That is useful for live used/free memory, but it is not authoritative for physical capacity or hardware reservations. This audit reads the GPU's own BAR0 registers through a kernel mapping and interprets them using NVIDIA's published register semantics.

2.2 Measurement chain

1. `ssdiag` 0.6.14 was built from source (`tools/ssdiag` in the upstream project).
2. It embeds the Intel-signed, hash-pinned `pmxdrv.sys`, loads it as a demand-start service for the probe, maps the GPU's physical BAR0, and removes the service on exit.
3. The local build was instrumented with one extra read: `NV_PFB_PRI_MMU_LOCAL_MEMORY_RANGE` at BAR0 offset `0x100ce0`. Upstream 0.6.14 does not print this register.
4. `nvidia-smi` was used only as a secondary cross-check of the live memory split and PCIe/power state.
5. The audit performed no GPU register writes. The tensor override was already live from the user's USB boot; this audit only read it.

The temporary kernel driver is visible to EDR/anti-cheat as a service install (SCM event 7045), even though it is removed on exit. That is a property of this measurement method, not of the card.

2.3 Register semantics

The interpretation was checked against NVIDIA's open-source kernel module source, not inferred from the driver's displayed numbers:

- `kmemsysReadUsableFbSize_GA102` reads `NV_USABLE_FB_SIZE_IN_MB` at `0x1183a4` and states: "VBIOS Devinit calculates the usable FB size calculated and published in NV_USABLE_FB_SIZE_IN_MB."
- `kmemsysReadUsableFbSize_GP102` reads `NV_PFB_PRI_MMU_LOCAL_MEMORY_RANGE` at `0x100ce0`, decodes `LOWER_MAG << (LOWER_SCALE + 20)`, and applies `15/16` only when the ECC bit is set.

That is why 16,384 MiB and 16,376 MiB are both correct: one is the raw MMU local-memory range, the other is the VBIOS-declared usable FB after its carveout.

3. Target identity

Property	Value
Marketing name	NVIDIA GeForce RTX 4080 SUPER
Architecture	Ada Lovelace, AD103
PCI vendor/device	`10de:2702`
Subsystem	`1569:f304`
PCI address	`0000:04:00.0`
VBIOS	`95.03.44.00.4e`
Board ID	`0x400`
GPU part number	`2702-400-A1`
CUDA compute capability	8.9
Active SMs	80
Active tensor cores	320
Memory bus	256-bit
Reported power limit	320 W (min 150 W, max 320 W)

The GPU is a display-class function on the Intel PCH root port `00:1c.4` (`8086:7a3c`), not on the CPU PEG port.

4. Raw register evidence

All values below are direct BAR0 reads from the audit run. The full dump is in `evidence/ssdiag-live.txt`.

Register	BAR0 offset	Raw value	Meaning
`NV_PFB_PRI_MMU_LOCAL_MEMORY_RANGE`	`0x100ce0`	`0x00000209`	`LOWER_MAG=32`, `LOWER_SCALE=9`, `ECC=0` → 16,384 MiB
`NV_USABLE_FB_SIZE_IN_MB`	`0x1183a4`	`0x00003ff8`	16,376 MiB usable FB
`WPR2_LO`	`0x1fa824`	`0x1ffffe00`	empty-window default
`WPR2_HI`	`0x1fa828`	`0x00000000`	no active WPR2 window
`MMU_LOCK_LO`	`0x1fa82c`	`0x1ffffff0`	empty lock window
`MMU_LOCK_HI`	`0x1fa830`	`0x00000000`	no active MMU lock window
`VGA_WORKSPACE`	`0x625f04`	`0x00000001`	VGA workspace flag set
`PSTRAPS_SET0`	`0x101000`	`0x00400080`	BAR straps
`PSTRAPS_SET1`	`0x10100c`	`0x02013000`	BAR straps
`FEAT PLM`	`0x823800`	`0xffffffff`	OPEN
`FEAT SS0`	`0x82381c`	`0x88888888`	all rate fields at full-rate value
`FEAT SS1`	`0x823820`	`0x00000008`	low nibble **8**
`FEAT READOUT_1`	`0x823818`	`0x00000000`	full speed, no throttle fields

Register	BAR0 offset	Raw value	Meaning
WPR_PLM	0x1fa7c4	0x0004cb8f	closed
WPR_CFG_PLM	0x1fa7cc	0x0004cb8f	closed
FBPA_PLM	0x9a0148	0xffffff0f	closed
MMU_PLM	0x1fa7c8	0x0045cb8f	closed
FUSE_OPT_FEATURE_OVR	0x8203f0	0x00000000	override path not blocked here
FUSE_VER	0x824140	0x00000001	fuse version 1
OPT_GPC_DISABLE	0x9a014c	0x00000000	no GPC-level disable fuse
OPT_GPC_DEFECTIVE	0x9a0168	0x00000000	no GPC-level defect fuse
DISABLE_FUSE_PROG	0x82007c	0xbadf5040	PRIV_VIOLATION (not readable at this level)

4.1 Decoding LOCAL_MEMORY_RANGE

```
raw          = 0x00000209
LOWER_MAG    = (raw >> 4) & 0x3f = 32
LOWER_SCALE  = raw & 0xf         = 9
ECC          = (raw >> 30) & 1   = 0
bytes        = 32 << (9 + 20)    = 0x400000000
             = 17,179,869,184 bytes
             = 16,384 MiB
```

Because ECC is off, the 15/16 correction does not apply.

4.2 Why 16,384 and 16,376 are both real

Value	Source	Interpretation
16,384 MiB	LOCAL_MEMORY_RANGE	raw MMU local-memory range
16,376 MiB	NV_USABLE_FB_SIZE_IN_MB	VBIOS-declared usable FB after its carveout
8 MiB	difference	VBIOS/firmware reserve

The VBIOS does not publish a "raw minus reserve" breakdown in these registers. The 8 MiB is a measured difference, not a named field.

5. VRAM accounting

Slice	MiB	GiB	% of raw	Nature
Raw GDDR6X population	16,384	16.000	100.000%	8 × 2 GiB packages
VBIOS usable-FB ceiling	16,376	15.992	99.951%	NV_USABLE_FB_SIZE_IN_MB
VBIOS/hardware reserve	8	0.0078	0.049%	not in the usable-FB register
Driver/WDDM reserved	329	0.321	2.008%	software reservation
Application-visible total	16,047	15.672	97.942%	16,376 − 329
Used at 17:52 snapshot	13,250	12.939	80.872%	desktop/apps, transient
Free at 17:52 snapshot	2,798	2.732	17.078%	transient

nvidia-smi reported 16376 / 329 / 13250 / 2798 MiB at the same time. The one-MiB rounding in 329 + 13250 + 2798 = 16377 is normal MiB rounding in NVAPI.

5.1 What the 8 MiB is — and is not

- Not WPR2. WPR2_HI = 0 means no window is active. WPR2_LO = 0x1ffffe00 is the documented empty-window default. Reading 0x1ffffe00 as "512 MiB locked" is a misread.
- Not ECC. ECC is off (LOCAL_MEMORY_RANGE ECC bit = 0), and NVAPI reports ECC N/A.
- Not a fixed 1 MiB VGA hole. The VGA workspace flag is set, but the withheld amount scales with capacity: 8 GiB field captures report 8,188 MiB (4 MiB withheld), 6 GiB captures report 6,141 MiB (3 MiB withheld), and this 16 GiB board reports 16,376 MiB (8 MiB withheld). That is exactly 1/2048 of raw capacity in each case.
- A capacity-scaled VBIOS/firmware carveout. The register only publishes the usable value; it does not name the consumer. The safe conclusion is: 8 MiB is unavailable as framebuffer and is not recoverable by the tensor unlock.

5.2 What the 329 MiB is

`nvidia-smi`'s "Reserved" is the WDDM/KMD reservation: driver internal allocations, page tables, display/scanout and protected structures. It is software, not physically dead DRAM. It can change with driver version, display topology and mode. It is not a hardware lock and cannot be reclaimed by a register write.

5.3 Is any VRAM "locked"?

Definition	Amount	Verdict
Physically absent	0 MiB beyond 16 GiB	No hidden capacity
VBIOS/hardware reserve	8 MiB	Real, not unlockable
Driver/WDDM reserve	329 MiB	Software accounting; not a fuse
BAR1 CPU aperture	256 MiB mapped of 16 GiB	Mapping window, not lost VRAM
WPR2 protected window	0 MiB active	Empty at audit time
ECC reserve	0 MiB	ECC off

The only true hardware reservation is 8 MiB. The 329 MiB is software; the 256 MiB BAR1 is an aperture.

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6. Tensor-rate state

The card is unlocked at the time of the audit:

```
FEAT PLM       = 0xffffffff   OPEN
FEAT SS0       = 0x88888888   all eight decoded rate fields = 0 (override + full rate)
FEAT SS1       = 0x00000008   low nibble = 8
FEAT READOUT_1 = 0x00000000   full speed
VERDICT [ada]: UNLOCKED (override values live)
```

The tool's `SS0` decode is `[0* 0* 0* 0* 0* 0* 0* 0*]`. In its field semantics, `0` means override plus full rate; locked states carry non-zero fractional fields. `READOUT_1 = 0` means none of the compute-throttle fields are set.

This is a rate-selector override, not an overclock and not a VRAM change:

- It removes the half-rate clamp on the FP32-accumulate tensor paths: FP16, BF16, TF32 and the FP8 path in the same family.
- FP16-with-FP16-accumulate already runs full rate and is a control (no gain).
- FP32 SIMT and the tested INT8/cuBLAS path are unaffected.
- The override is volatile across a full power loss. Sleep/FLR behavior varies by platform; a cold boot requires replaying the USB stage.

6.1 What the rate actually is

- Active tensor hardware: 80 SM × 4 tensor cores = 320 tensor cores.
- Unlocked FP16/BF16 dense rate law: 1,024 FLOP/SM/clk for the FP32-accumulate path; locked is 512 FLOP/SM/clk.
- At the card's 2.55 GHz rated boost: `80 × 1024 × 2.55 ≈ 208.9 TFLOPS` FP16/BF16 dense.
- The published same-die `2702` measurement is 101.0 TFLOPS locked → 202.7 TFLOPS unlocked (2.01×) for FP16/BF16 FP32-accumulate, with the FP16-accumulate control flat at 1.00×. The live selector read matches the unlocked state.
- At the 320 W board limit, sustained cuBLAS gains are typically lower (about 1.66–1.78× on 4080 SUPER reports) because the doubled toggle rate reaches the power wall. The 2.01× figure is the short-burst/raw-pipeline ceiling.

No fresh CUDA GEMM was run in this audit. The 202.7 TFLOPS number is a same-die reference measurement; the live `SS0/SS1/READOUT_1` values are direct hardware reads from this card.

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7. Tensor hardware and TPC floorsweep

`ssdiag` read the per-GPC effective TPC masks (`STATUS`, bit n = TPC n of that GPC):

```
GPC0 6/6   GPC1 6/6   GPC2 6/6   GPC3 4/6 (TPC4,TPC5 off)
GPC4 6/6   GPC5 6/6   GPC6 6/6   GPC7 0/6 (unused eighth slot)
```

Item	Count
Active TPCs	40
Active SMs	80
Active tensor cores	320
GPC3 disabled TPCs	2
GPC3 disabled SMs	4
GPC3 disabled tensor cores	16

`OPT_GPC_DISABLE = 0` and `OPT_GPC_DEFECTIVE = 0`. The tool reports no defect bit on the disabled TPCs and labels them "override candidates (write path out of scope)."

7.1 Interpreting GPC7 and the 8-TPC verdict

The tool's generic verdict says "8 TPCs fused off in 2 GPCs" because it counts the all-off GPC7 slot. AD103 is a 7-GPC/80-SM-class die; GPC7 is the unused eighth register slot, not 12 recoverable SMs. The real binning cut is the 2 TPCs in GPC3.

This is die binning, not the tensor-rate lock. There is no supported or validated write path to turn those 2 TPCs on. Treat any such attempt as experimental and potentially destabilizing; it is not part of this unlock.

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8. Platform limits

8.1 BAR1 / Resizable BAR

Config space reports:

```
BAR1 supported [16G/8G/4G/2G/1G/512M/256M/128M/64M] current 256 MiB (size_idx 8)
```

Only 256 MiB of the 16 GiB is mapped into the CPU's PCIe BAR window. The card supports a 16 GiB BAR1; the current 256 MiB setting is a firmware/BIOS configuration (Above 4G Decoding / Re-Size BAR). Enabling ReBAR would make the full VRAM CPU-visible, but it would not add a byte of VRAM or change the 16,376 MiB usable ceiling.

8.2 PCIe link

```
LnkCap max Gen4 x16 | LnkSta Gen1 x4 (downgraded)
```

The GPU is on Intel PCH root port `00:1c.4` (`8086:7a3c`), not the CPU PEG port. At idle the link drops to Gen1 (normal power saving); the x4 width is the real limitation. The card can do Gen4 x16, but this slot/riser is wired x4. That is a platform bandwidth bottleneck, not a GPU lock. Moving the card to the CPU x16 slot, or replacing an x4 riser/adapter, is the fix.

8.3 Power limit

NVAPI reports 320 W current/default, min 150 W, max 320 W. This is a VBIOS power-table cap, not a hardware fuse. It limits sustained tensor gains; changing it requires a VBIOS/power-table modification, not the tensor unlock.

8.4 ECC and MIG

ECC is off (`ECC=0` in `LOCAL_MEMORY_RANGE`; NVAPI N/A). MIG is N/A on this GeForce SKU. Neither consumes a hidden VRAM pool here.

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9. What can and cannot be unlocked

Item	Status	Can it be unlocked?
Tensor-rate selectors	Open (`PLM=0xffffffff`, `SS0=0x88888888`, `SS1=0x8`)	Already unlocked; volatile, replay after cold boot
VBIOS 8 MiB reserve	Withheld	No; outside the usable-FB ceiling
WDDM 329 MiB reserve	Software	No register unlock; may change with driver/display configuration
BAR1 256 MiB aperture	Firmware setting	Yes, via BIOS ReBAR; does not add VRAM
GPC3 TPC4/TPC5	Binning	No supported path; experimental only
GPC7	Unpopulated slot	No; not 12 real SMs
320 W power cap	VBIOS	VBIOS/power-table modification only
PCIe x4	Platform wiring	Move to CPU x16 slot or replace riser
WPR/FBPA/MMU PLMs	Closed	Not opened by the tensor unlock; no capacity gain
ECC reserve	Off	No reserve to reclaim
MIG	N/A	Not applicable

9.1 Can more VRAM be unlocked?

No. The physical population is 16 GiB (8 × 2 GiB GDDR6X). The card exposes 16,376 MiB as usable FB and 16,047 MiB to applications after the driver reserve. The 8 MiB is a VBIOS/firmware carveout; the 329 MiB is a driver reservation. There is no 24 GB/32 GB array behind a fuse, and no register in this dump represents an additional capacity pool.

The "32 GB RTX 4080 SUPER" cards in circulation are physical mods with different memory packages, not a software unlock.

10. Evidence quality and limitations

Claim	Evidence	Confidence
Raw 16,384 MiB / usable 16,376 MiB	BAR0 MMIO + NVIDIA register semantics	High
8 MiB VBIOS reserve	Difference of two hardware registers	High for amount; medium for purpose
329 MiB driver reserve	NVAPI snapshot	Medium (software accounting)
Tensor override live	BAR0 `PLM/SS0/SS1/READOUT_1`	High
40 TPC / 80 SM / 320 tensor cores	BAR0 TPC map + CUDA SM count	High raw; medium interpretation
PCIe x4	PCI config space `LnkSta`	High
320 W limit	NVAPI	Medium (secondary)
Used/free memory	NVAPI snapshot	Low/transient

10.1 Limitations

- The 8 MiB VBIOS reserve is measured as a difference; the register does not name its consumer. The 1/2048 pattern across 6/8/16 GiB field captures is strong evidence of a capacity-scaled firmware carveout, but the exact purpose is not documented in these registers.
- The TPC interpretation uses the AD103 7-GPC/80-SM topology and the live 80-SM CUDA report. The raw `STATUS` words are hardware reads; the "GPC7 is unpopulated" reading is an inference from the SKU topology, not a separate fuse read.
- No fresh CUDA GEMM was run. The 202.7 TFLOPS figure is a same-die reference measurement; the live selector values are direct hardware reads.
- Current used/free numbers are a transient desktop snapshot. Close applications for a clean free-VRAM measurement.
- The audit did not test memory integrity, ECC, overclocking, or TPC enablement.
- `ssdiag` deploys a signed kernel driver for the duration of the probe. It is removed on exit; the SCM 7045 event remains in the log. EDR/anti-cheat may notice it.

11. Reproduction

The raw read is in evidence/ssdiag-live.txt. The added register read is:

```
BAR0 + 0x100ce0   NV_PFB_PRI_MMU_LOCAL_MEMORY_RANGE
                  LOWER_MAG   = (v >> 4) & 0x3f
                  LOWER_SCALE = v & 0xf
                  ECC         = (v >> 30) & 1
                  bytes       = (LOWER_MAG << (LOWER_SCALE + 20))
                  if ECC: bytes = bytes / 16 * 15
```

FB_SIZE_MIB is BAR0 + 0x1183a4, low 16 bits, in MiB. The FEAT cluster is 0x823800/0x823818/0x82381c/0x823820. The TPC map is CTRL = 0x820838 + 4*g, STATUS = 0x820c38 + 4*g.

Recommended sequence:

1. ssdiag --dry-run — confirm the GPU is enumerated and the platform gates.
2. Run ssdiag elevated — it loads the driver, reads BAR0, and removes the service.
3. Keep the raw output with the audit record.
4. Do not write SS0/SS1 unless you are intentionally re-applying the tensor override.
5. Do not write TPC CTRL registers; no supported path exists.

12. Conclusion

The card is a 16 GiB RTX 4080 SUPER with 16,376 MiB of hardware-usable framebuffer. The only true hardware reservation is 8 MiB (0.049%), a capacity-scaled VBIOS/firmware carveout. The 329 MiB shown by NVAPI is a software/driver reservation, and the 256 MiB BAR1 is a CPU aperture, not lost VRAM.

The tensor-rate lock is currently removed and the selectors are at full rate. The 2 disabled TPCs in GPC3 are binning, not the tensor-rate limiter. The meaningful remaining bottlenecks are the x4 PCIe link, the 256 MiB BAR1 window, and the 320 W VBIOS power cap.

There is no additional VRAM to unlock. The physical memory does not exist beyond 16 GiB; the 8 MiB VBIOS reserve and 329 MiB driver reserve are not a hidden capacity pool.

Appendix A — evidence files

File	SHA-256
evidence/ssdiag-live.txt	eb5faf9583bd9692d5bb74009b4f3d003ea25117b75d9e08237b702a67c8f0d2
evidence/hardware-snapshot.txt	bf262278e89cdc7926b78a654f7992b7a92df13a864d8d59b883c20ec3fbcca4

The raw log includes the platform gates, the GPU instance path, BAR0 ranges, the core/debug block, the FEAT clusters, the Ada fuse/TPC block, the per-GPC TPC floorsweep, the PCIe/gen2 probe, the FEAT sweep, the PLM survey and the final verdict.

Appendix B — document history

Version	Date	Change
1.0	2026-11-07	Initial hardware audit for the local RTX 4080 SUPER

This report is a read-only hardware audit. It is not a warranty, a stress-test qualification, or an endorsement of TPC/fuse modification. The tensor unlock remains volatile and must be replayed after a cold boot.